

DEVELOPMENT OF LINE OF THERAPY RULES IN MULTIPLE MYELOMA: OPTUM CLAIMS

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1. AMENDMENTS AND UPDATES

Amendment or update no	Date	Section of study protocol	Amendment or update	Reason

2. BACKGROUND AND RATIONALE

There is an urgent need for high-quality, timely real-world evidence (RWE) to enrich our understanding of multiple myeloma (MM) patient populations in the routine care setting, to inform R&D decision-making, and to drive evidence-based development strategies across the GSK enterprise. RWE is generated using real-world data (RWD), which are data on patient health status, or the delivery of health care routinely collected from a variety of sources, including but not limited to, electronic health records, health claims data and disease registries. A first step in developing an above asset MM cohort is to develop a line of therapy (LOT) algorithm, which will be implemented based on the receipt and timing of therapy because LOTs are not explicitly captured in claims data. This methods document describes development of the LOT algorithm and how it can be implemented within a health claims setting. A next step will be to generate a standing cohort (i.e. real-world analytical product [RWDP]) and will be described in a future study.

2.1. Overview of Multiple Myeloma

MM is a clonal proliferation of abnormal plasma cells in bone marrow, and is the second most common hematological malignancy, accounting for approximately 10% of all cases

(Rajkumar, 2022). MM is a relatively rare disease with 36,110 incident cases and 12,030 deaths in the United States (US) estimated in 2025 (Siegel, 2025).

Treatment of MM is initiated when patients are symptomatic, or have active disease based on the International Myeloma Working Group (IMWG) definition (Rajkumar, 2014). Standard frontline therapy is decided by chromosomal abnormalities by FISH (fluorescence in situ hybridization), p53 mutation analysis by DNA sequencing, patient comorbidities, and eligibility for stem-cell transplantation (SCT) (Rajkumar, 2020a; Rajkumar, 2020b). For transplant-ineligible patients, the preferred first-line treatment regimens are daratumumab, lenalidomide, or dexamethasone-based regimens; other regimens are recommended in certain circumstances (Rajkumar, 2020b; Cowan, 2022). With an enhanced understanding of the pathophysiology and disease heterogeneity, and the advent of novel treatments, there have been improvements in overall survival of MM patients (Cowan, 2022; LeBlanc, 2024).

Despite the positive trends in disease survival and decrease in mortality (Jiang, 2025), the MM inevitably results in relapse after initial therapy (Kastritis, 2022; Mohy, 2019). The choice of treatment from various available regimens to treat relapsed and refractory MM (RRMM) is based on numerous factors. There are no firm treatment guidelines as clinical decisions are based on potential toxicities, patient and physician preferences, prior treatment, cytogenetic risk, comorbidities, myeloma markers and financial considerations (Kumar, 2016; Rajkumar, 2012; Laubach, 2016). Treatment options for RRMM patients include re-administration of initial agents for those who are naïve or sensitive to the novel drug combinations and were not exposed to them in the previous six months. Patients who are refractory to the novel drugs should be considered for regimens that do not contain the drugs that they progressed on. Other considerations for RRMM include newer agents in the same class (i.e. carfilzomib and pomalidomide), novel mechanisms of action drugs (XPO inhibitors, BCL-2 inhibitors, cereblon E3 ligase modulators, and T-cell engaging therapies such as bispecific antibodies or CAR-T cell therapy), combination therapies and autologous/allogeneic stem-cell transplants (SCTs) (NB: allogeneic SCTs are rarely done) (Hernández-Rivas, 2022). The complexity of the MM treatment landscape, due to multiple treatment phases, gaps between regimens, and combination therapies of three or more

drugs, has made it increasingly difficult to identify the real-world clinical pathways of the patients (Hernández-Rivas, 2022; Laubach, 2014; Ocio, 2014).

Identifying the prior lines of therapies (LOT) can support selection of treatment regimens for relapsed/refractory MM patients, determine eligibility for enrolment into clinical trials, support adherence, and enable accurate health technology assessment for approvals (e.g., FDA) (Cardoso, 2020; Saini, 2021). As additional novel therapies for MM become available, particularly in the second and third lines, there is a need to understand various lines of therapy for RRMM patients. The goal of this project is to develop a set of rules to identify lines of therapy for MM patients using administrative claims data. LOT rules will be validated with clinical adjudication and by comparing identified treatment sequences to established external benchmarks and published LOT definitions from registries, claims databases and pivotal trials where available (NCCN, 2026; Rajkumar, 2015).

2.2. Multiple Myeloma LOT Algorithm

This document describes an updated version of the MM LOT algorithm rules that GSK and Optum worked to develop (Hanna et al.; GSK Project 219869, Jan 2025). Optum implemented previous versions of the LOT algorithm, for which the output of the first iteration of the MM LOT algorithm was analysed using Optum Market Clarity data and presented to GSK's external experts (EES). EE feedback led to revisions in the first version of the LOT algorithm, principally, but not limited to, exclusion of corticosteroids by themselves as an MM treatment.

Optum Market Clarity data was used for the original LOT build by Optum with the future intent of using the linked EHR data to validate the LOT algorithm in claims. It was however found that difference in data granularity and temporality of the claims versus the EHR data made validation not plausible. Feasibility results, based on Market Clarity data, are presented in Appendix A. A table summarizing the demographic profile of the original patient sample in Market Clarity, provided by Optum, is presented in Appendix B.

Output from this study will be a LOT algorithm built on the foundations of the prior work between Optum and GSK. The LOT algorithm will be applied to a MM US claims cohort to allow rapid identification and characterisation of MM by LOT, per team needs.

3. STUDY PURPOSE AND OBJECTIVE(S)

The purpose of this study is to develop and validate a LOT algorithm for MM to be used for administrative insurance claims data. The objective of this project is to:

- 1) Develop a LOT algorithm to identify up to five LOTs for MM using administrative claims data.

In addition, sensitivity analyses will be conducted to further refine the algorithm ensuring it is fit-for-purpose. This is a methods document; study-style language is retained for clarity and is illustrative only and should not be taken as an approved study design.

4. STUDY DESIGN AND PLAN

4.1. Study Design

This study will use Optum's Clinformatics Data Mart (CDM) to identify newly diagnosed MM patients during the period of 1 January 2016 up to the latest data available (identification period) to reflect more recent patient populations and treatments. The date of the first claim with a diagnosis code for MM during the identification period will be defined as the index date. All patients are required to have a baseline of at least 6 months before the index date. The variable-length follow-up begins on the index date and ends at the earliest of death, health plan disenrollment or the end of the study period. Refer to Section 4.3 for further information.

Starting the study period in 2015 and extending up to the latest data available will provide a sufficiently long duration of data for patients with index dates earlier in the identification period to capture patients who are newly diagnosed and develop relapsed/refractory MM.

4.2. Eligibility Criteria

The final study population for the LOT dataset will be exclusively defined by the inclusion and exclusion criteria listed below. While the specific application of these criteria may involve initial study team discretion to refine and optimize cohort selection, the ultimate LOT dataset will strictly comprise only those patients who meet these precisely specified inclusion and exclusion criteria. The minimum mandatory requirement for cohort membership is that **patients must have at least one inpatient medical claim or 2 outpatient medical claims for MM in any position on the claim during the study period and no MM oncology therapy use during the baseline period.** Criteria to be explored as sensitivities are detailed in 4.2.1.

Inclusion Criteria

Patients are eligible for the project sample if they meet the inclusion criteria listed below.

- 1. **MM diagnosis:** At least one inpatient medical claim with a diagnosis code for MM in any position (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or ≥ 2 medical claims for MM in any position on the claim, on separate days within 90 days, during the identification period. The index date is the date of the first medical claim for MM within the identification period. Diagnosis codes for MM are in the programming specification document.

As data in Optum CDM is collected primarily for health insurance and not research purposes, all MM diagnosis codes are to be considered and the code list will not be limited to those specifically indicated for newly diagnosed MM.

- 2. **Adults:** ≥ 18 years in index calendar year.

3. Baseline continuous enrollment (CE) of at least six months: At least six months of CE with medical and pharmacy benefits before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.

The purpose of the baseline is to create a period during which patients who were previously diagnosed with MM are identified; if treated in the baseline period, these patients will be excluded (see Exclusion Criteria below). If not treated in the baseline period, these patients will be assumed to be smoldering until the point of treatment in the follow up period.

4. Follow-up CE of at least 1 day: ≥ 1 day of CE starting on the index date until death or study period end. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.

o As noted in Section 4.5, mortality data includes month and year of death. The assigned date of death for dataset construction is set to the 15th of the month.

5. Evidence of treatment with agents for the purpose of treating MM based on FDA-approval and NCCN guidelines during the follow-up period:

Patients will be required to have evidence of an approved MM oncology therapy in claims during the follow-up period. These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.

Receipt of corticosteroids only (e.g., dexamethasone) will not be considered evidence of receiving an approved MM oncology therapy. NDC codes and HCPCS codes for approved MM therapies are in the programming specification document, tabs "MM approved therapy - NDC" and "MM approved therapy - HCPCS", respectively.

Exclusion Criteria

Patients meeting the following criteria are flagged for potential exclusion from the study sample.

1. Evidence of treatment with agents for the purpose of treating MM based on FDA-approval and NCCN guidelines during the follow-up period:

Patients will be required to have evidence of an approved MM oncology therapy in claims during the follow-up period. These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.

4.2.1. Exploratory Eligibility Criteria

The following specifies potential inclusion and exclusion criteria that may be applied on an exploratory basis, in addition to or in place of the stated eligibility criteria, to improve the accuracy and consistency of LOT classification.

Inclusion Criteria

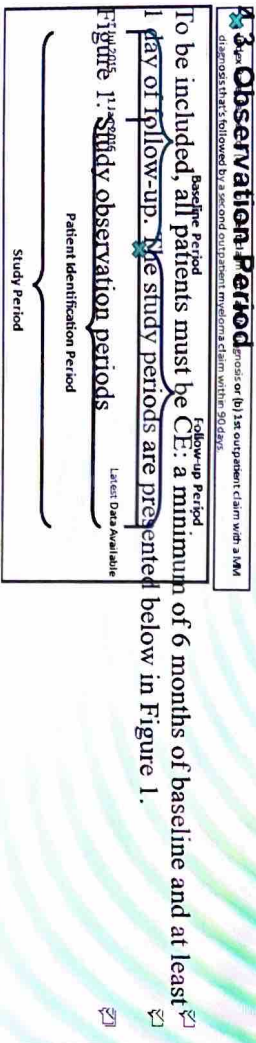
1. MM diagnosis: Instead of the usual requirement for ≥ 2 MM medical claims on separate days within a 90-day window, this exploratory criterion evaluates shorter windows of 60 and 30 days. Specifically, patients must have at least one inpatient medical claim with a diagnosis code for MM in any position (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or ≥ 2 medical claims for MM in any position on the claim, on separate days within 60 or 30 days during the identification period. The index date is the date of the first medical claim for MM within the identification period. Diagnosis codes for MM are in the programming specification document.

2. CE Follow-up Requirement: Patients with ≥ 3 months of CE (or have an assigned date of death < 3 months after the index date), without gaps in their enrollment, measured from index date will be flagged for consideration for sensitivity or secondary analyses. The follow up period ends on the earliest of health plan disenrollment, study period end or date of death, whichever comes first.

o For patients with an assigned date of death < 3 months after the index date, these patients are retained and have a limited follow-up period that ends at the assigned date of death.

Exclusion Criteria

- 1. Evidence of an MM oncology therapy during the baseline period: ≥ 1 medical or pharmacy claim for an FDA-approved MM oncology therapy during the baseline period, to ensure that treatment exposure and LOT assignments reflect incident therapy starts at the index date and are not confounded by ongoing or recent prior MM treatments. The MM oncology therapies are listed in the programming specification document.
- 2. Evidence of another cancer in the baseline period: Patients with either ≥ 1 inpatient or ≥ 2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer will be excluded (patients with ≥ 2 outpatient claims on separate days within 60 or 90 days apart will also be flagged for consideration). Codes (ICD-9-CM and ICD-10-CM) are in the programming specification document.
- 3. Evidence of pregnancy: ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the baseline or follow-up period. Codes for pregnancy are in the programming specification document.
- 4. Clinical trial participation: Evidence of clinical trial participation during the study period will be flagged. Codes for clinical trial participation are in the programming specification document.



4.4. Data Sources and Collection

Data for this study were obtained from the Optum Clinformatics® Data Mart (CDM), a HIPAA-compliant, de-identified administrative claims database. The CDM links enrolment and provider information with medical, facility, physician, and pharmacy claims, as well as outpatient laboratory results from major national labs. It includes service dates from January 2007 to the present, covers approximately 15–20 million lives annually across all 50 states, and restricts membership to individuals with both medical and pharmacy benefits.

4.5. Mortality Data

Data on mortality in the Optum CDM is captured from various sources to provide the most complete information on death available. As noted in Section 4.1, assigned date of death was used as a criterion to end the variable follow-up period.

- 1. Centers for Medicare and Medicaid Services (CMS); Mortality information derived from CMS data used in support of Medicare services

2. Social Security Administration Death Master File (DMF)
3. Facility claims where discharge status = expired
4. Member coverage, reason for coverage discontinuation = death
5. Electronic medical records (EMR)
6. Obituary data, externally sourced

5. LINE OF THERAPY ALGORITHM

This section describes the algorithm development process including identifying LOTs and how the variables in the algorithm will be derived. The variables and variable definitions for the LOT algorithm were informed by the clinical literature, input from Optum including an expert hematologic oncologist consultant, and this study team.

For each patient, up to five lines of therapy will be identified for the entire follow-up period. The LOT variables are described immediately below and the analytic methods for the LOTs, including the LOT validation, are described in Section 5.3.1.

The MM therapies will be identified from procedure codes (e.g., infusions) in medical settings, including facility outpatient and ambulatory care settings, inpatient hospitalizations, and NDC codes in prescriptions recorded in claims.

The medications and associated codes used in the LOT algorithm are in the programming specification document. All rules described below are based on oncology therapies only (e.g., not corticosteroids).

5.1. LOT Inputs

- **MM therapies:** Approved MM oncology therapies as identified from the FDA, NCCN guidelines (NCCN, 2026), and peer-reviewed literature (Rajkumar, 2015). NDC codes

and HCPCS codes for approved MM therapies are in the programming specification document.

- The following therapy is excluded from LOTs: mesna (treatment of MM to protect the patient's bladder against the toxic side-effects of certain anticancer drugs).

- **Stem cell transplant (SCT):** A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as ≥ 1 procedure code for an SCT.

- **Planned Tandem SCT:** Patients will be classified as having a planned tandem SCT if they receive at least two autologous SCTs ≥ 60 to ≤ 180 days apart; otherwise, they will be considered unplanned (Rajkumar, 2015). An autologous SCT followed by an allogeneic SCT will not be considered planned because an allogeneic SCT typically follows failed autologous SCTs. Presence of an unplanned SCT will end the current LOT the day before the unplanned SCT if the LOT has not otherwise ended prior to the date of the unplanned SCT.

- **Allogeneic SCT:** Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT.

- **Maintenance regimen:** A maintenance regimen is a period of 120 days or longer during which only a valid maintenance therapy is available (see list below of valid mono/dual maintenance therapies). Maintenance is not required to be prescribed continuously for the duration of the 120 day period. Furthermore, the maintenance therapy must have been a part of the LOT's initial regimen (for LOT1 the initial regimen includes all MM therapies identified during the first 60 days of LOT1, for LOT2-5 the initial regimen is defined as all MM therapies identified during the first 30 days of the LOT). A maintenance regimen may occur in one of two situations during a LOT:

1. The LOT's initial regimen transitions into a maintenance regimen, such that non-maintenance medications present during the initial regimen are discontinued leaving only maintenance medications. For example, a patient has an initial regimen of

cyclophosphamide and daratumumab, and then discontinues cyclophosphamide leaving only daratumumab, which is a valid mono maintenance per the list below.

2. Following an autologous SCT (within 180 days of the SCT or in the case of a tandem autologous SCT within 180 days of the second SCT). The expectation after an SCT is for a maintenance period and thus instead of the usual 120 days, we check for a valid 30-day (minimum) maintenance period. Any valid maintenance regimens (see list below) present after a single autologous SCT or after the second SCT of a tandem autologous SCT will be considered. These maintenance drugs must have been part of the LOT's initial regimen.

a. If a full 30-day period is unable to be observed due to the end of follow-up, whatever days are available will be the maintenance period to avoid a new LOT starting immediately before the end of follow-up.

The following will be considered maintenance therapies, per NCCN Guidelines (Kumar, 2020).

- Lenalidomide monotherapy (NCCN Preferred Regimen for both transplant and non-transplant candidates)
- Bortezomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Daratumumab monotherapy (NCCN Other Recommended Regimen for transplant candidates)
- Ixazomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Thalidomide monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Bortezomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances, for non-transplant candidates)

- Carfilzomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances for transplant candidates)

- Daratumumab/lenalidomide dual therapy (NCCN Useful in Certain Circumstances for transplant candidates)

Analysis of real-world data consistently shows that patients may receive regimens that are inconsistent with NCCN guidelines. For example, although daratumumab is recommended for transplant candidates only, it is possible that patients who do not undergo an SCT will receive this medication. Therefore, any of the maintenance medications listed above will be considered maintenance medications for all patients, regardless of transplant status.

Once a maintenance regimen has begun (e.g., observing a valid maintenance medication(s) by itself for at least a 120 day period), it continues until discontinuation (defined below), presence of an SCT, right censoring (i.e. end of follow-up), or the addition of any non-maintenance MM therapy automatically causes the maintenance regimen to end.

5.1.1. LOT Variables and Parameters

- LOT start date and regimen

- First-line (LOT1):

- LOT1 start date: LOT1 begins on the date of the first fill for an MM therapy following a patient's index date.

- LOT1 regimen: The LOT1 regimen includes all MM therapies received within 60 days (Madduri, 2021) on and following the LOT1 start date. For LOT1, the first therapy received is considered the induction therapy.

- Second-line and later LOTs (LOT2-LOT5):

- LOT2-LOT5 start date: Each subsequent LOT (LOT2-LOT5) begins on the earliest of:

- A stem cell transplant (SCT), allogeneic or an unplanned autologous SCT),
 - CAR-T cellular therapy, or
 - The date of the first administration for a new MM agent that was not part of the previous LOT regimen.
- **LOT2-LOT5 regimen:** For LOT2-LOT5, the LOT regimen includes all MM therapies received within 30 days on and following the LOT start date.

• LOT Classification

- **First-line (LOT1):** The first line of therapy received by patients with newly diagnosed MM during the identification period. This includes induction therapy and may also include maintenance and SCTs.
- **Second-line (LOT2) and later LOTs (LOT3-LOT5):** Defined by initiation of a new LOT after the end of the previous LOT, as determined by the LOT start and end rules above (e.g., initiation of a new MM agent not present in the previous LOT regimen, CAR-T, or an SCT that ends the prior LOT). For LOT2 and later LOTs, the regimen identification window is 30 days (as specified under LOT regimen).
- **CAR-T:** CAR-T cellular therapy infusions are classified as their own LOT. Therapies (including supportive agents, e.g., corticosteroids) given within 30 days of CAR-T cellular therapy are consolidated as part of the CAR-T LOT.
- **Allogeneic SCT:** Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.
- **LOT end date:** The LOT will continue until the earliest of any of the following:
 1. **Substitution** of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration.

2. Discontinuation of all agents in the regimen (with or without switch to a new agent):

Discontinuation of the regimen occurs when all MM agents in the LOT are stopped, either:

- Without initiation of a new MM agent (i.e., all agents in the regimen have reached their run-out dates and no further MM administrations/dispenses are observed), or
- With initiation of a new MM agent that was not present in the induction regimen (i.e., the regimen is interrupted by a new agent).

In both cases

- The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available (not simply the last administration date for infused/injected drugs or last dispense date for oral drugs).
 - If the current LOT is immediately interrupted by a new agent (the new agent starts on or before the run-out date of the current regimen), the LOT end date is the day before the first administration/dispense date of the new agent.
 - If the regimen is discontinued and no new agent starts before a qualifying gap, the LOT end date is the run-out date immediately before that gap.
 - The run-out date is calculated using a patient's combined infused/injected drugs (medical benefit) and oral drugs (pharmacy benefit). Overlapping pharmacy benefit prescriptions "push out" (extend) the pharmacy run-out date according to the LOT1 specifications.
 - The days' supply on medical claims is always missing and therefore a 28-day supply is assumed. When pharmacy claims are missing days' supply, or when days' supply values are anomalous, a 28-day supply is assumed.
3. **Unplanned SCTs:** If an autologous SCTs occurs > 180 days following a previous autologous SCT in the LOT, then the LOT will end the day before the latter SCT.
- Any allogeneic SCTs are considered a new LOT and the current LOT will end the

day before an allogeneic SCT occurs. Single autologous SCTs and tandem SCTs are not unplanned SCTs and are considered a continuation of the line of therapy.

4. **SCTs not followed by maintenance:** If an SCT is not followed by a maintenance regimen within 180 days, then the last day of the LOT is the date of the SCT.

5. **Death:** Ends the LOT

6. **Health plan disenrollment:** Ends the LOT

7. **End of the study period:** Ends the LOT

8. **End of maintenance regimen:** The LOT will end after a maintenance regimen ends due to:

- The introduction of any MM therapies including therapies that were part of the original regimen does not advance the LOT (e.g., a LOT has an initial regimen of Bortezomib+Lenalidomide and later the patient has a Bortezomib mono maintenance regimen; if Lenalidomide restarts at some point then the maintenance regimen will end as we observed 120+ days of only Bortezomib and the reintroduction of Lenalidomide could indicate progression).

- Unplanned SCTs (defined above), death, health plan disenrollment, and end of study period.

Events that will **not** end the LOT are:

- **Single autologous SCT or tandem stem-cell transplants (SCT) $\leq$ 180 days apart:**

SCTs are usually preceded by induction therapy and followed by consolidation and maintenance therapy. Tandem SCTs are two SCTs $\geq$ 60 to $\leq$ 180 days apart. These are considered planned and a continuation of the line of therapy. Codes for SCTs are in the programming specification document.

SCTs must be within 60 days after the end of the induction or maintenance regimen to be considered a continuation of the LOT. This 60-day rule was recommended by Optum's KOL¹ and informed by Charalampous et al.'s findings that the median time

¹ Optum's KOL is Dr Allan Stawis.

from end of induction to transplant was 33 days (interquartile range=27-42) (Charalampous, 2023).

- **The start of a maintenance regimen:** Receipt of a valid maintenance regimen (see 5.1. for definition)

5.2. Limitations

- The LOT algorithm is restricted to therapies approved for MM as specified in the current NCCN Guidelines and therefore will not identify off-label treatments, investigational agents, drugs used for other malignancies, or non-oncology medications with anti-cancer effects. This restriction may lead to under-ascertainment or misclassification of treatment exposures and LOT assignment, potentially biasing estimates of treatment patterns, duration, and outcomes and limiting generalizability to settings where non-NCCN or investigational therapies are used.
- Treatment patterns observed in Optum CDM reflect real-world practice within an insured population and may be influenced by formulary restrictions, access to care, provider practice patterns, and calendar time effects.
- Although LOTs are not explicitly captured in claims data, the algorithms used to identify LOT in this study are based on multiple clinically valid assumptions.
- Although prescription fills are available, such data do not reveal when or if the patient actually took the prescribed medication; thus, LOT will be identified based on claims for filled prescriptions.
- Administrative claims data are collected for the purpose of billing and reimbursement, not for coordinating medical care or conducting outcomes research. The data are subject to coding errors and underreporting of clinical conditions/events that do not trigger a billable event. Medications with blocked information (e.g., during the time that health insurers are identifying payment and authorization processes) may be missing from Optum's CDM.

- Treatment gaps used to define line transitions may reflect non-clinical factors such as administrative delays, scheduling issues, temporary treatment holds, or gaps in insurance coverage, rather than true disease progression or therapy failure.
- Maintenance therapy may be difficult to distinguish from continuation of active treatment, potentially leading to inconsistent line definitions across patients.
- As the LOT algorithm is static it could become outdated over time so will need to be periodically refreshed to reflect the current standard of care.

5.3. Data analysis

Overview Analyses will be descriptive and focused on (1) development and performance assessment of the claims-based LOT algorithm, (2) describe real-world treatment pathways/sequencing, and (3) sensitivity and subgroup analyses.

1. The LOT algorithm will be implemented to identify up to five sequential LOTs per patient using the rules described in Section 5 and the programming specification document.

- For each LOT, the analysis datasets will include LOT start and end dates, agents/regimens present, and indicators for receipt of CAR-T, type of SCT, and maintenance therapy, as appropriate.
- The distribution of number of LOTs per patient and time between LOTs (mean, standard deviation, median, IQR, min, max) will be tabulated. The most frequent regimens observed in LOT1, LOT2 and subsequent LOTs will be reported with counts and percentages.

2. Descriptive analyses of treatment pathways

- Treatment pathways will be described using sequence summaries and visualizations (e.g., Sankey plots) showing transitions across regimen classes and LOTs.
- Utilization of key interventions (SCT, CAR-T) and use of maintenance therapy will be reported as counts (n, %) and descriptively (median [IQR], mean [SD]).

3. Subgroup and sensitivity analyses

- Sensitivity analyses will assess robustness of LOT rules by varying key algorithm parameters (e.g., required amount of continuous enrollment in the follow up period, look-back windows for prior exposure) and by applying alternative cohort definitions (e.g., applying varying patient identification criterion, stricter continuous enrollment criteria, varying eligibility criteria to the starting patient population [as listed in Section 4.2.1]). Effects on primary outputs (number of LOTs, timing, regimen classification) will be reported.

5.3.1. Quality Control and Quality Assurance

Validation of the LOT algorithm will include clinical adjudication and comparing the finalized LOT to external benchmarks. These benchmarks include claims-derived treatment sequences, regimen frequencies and LOT distributions. Outputs will be compared to published LOT definitions and registry data, where available. Descriptive comparisons and standardized differences will be reviewed by the study team.

ANNEX - START-UP DOCUMENT TABLES

Appendix A. Original Feasibility in Market Clarity

These counts were generated by Optum during the proposal process from the criteria outlined in the original start-up document, and are for information purposes only. This feasibility had a study period was from 1 July 2015 to 30 September 2022, with an identification period for patients with newly diagnosed multiple myeloma from 1 January 2016 to 31 March 2022.

Criteria	Number of patients
Inclusion	
Patients with at least 1 non-diagnostic claim for multiple myeloma (ICD-9-CM=203.00 or ICD-10-CM=C90.00) during 1 January 2016 to 31 December 2021 (the identification period). The first date is the index date	95,796
Continuous enrollment: 6 months before index date (Baseline period)	35,938
Continuous enrollment: ≥3 months after index date, starting on index date (Follow-up period)	32,945
Age ≥18 years as of the index date	32,825
At least 1 claim for systemic MM cancer therapy on or after the index date. The date of the first claim is the first line of therapy start date	18,174
No evidence of systemic cancer therapy in baseline period	11,958
No evidence of MM in baseline period	7,971
No evidence of R/R MM (≥2 claims on different dates of service), in baseline period	7,864
No evidence of other cancers in baseline period	5,911

Counts among 5,911 patient population	
Patients with information on prescriptions written/administered in EHR	
Patients with at least 1 non-missing prescriptions written or medications administered occurring on or after LOT1 start date	2,467
Patients with at least 5 non-missing prescriptions written or medications administered on different dates of service occurring on or after LOT1 start date	1,853
Patients with physician description of treatment rationale in EHR	
Patients with at least 1 non-missing discontinuation reason (e.g., "lack of efficacy", "side effect", "alternate therapy prescribed") occurring on or after LOT1 start date	1,685
Patients with at least 2 non-missing discontinuation reasons (e.g., "lack of efficacy", "side effect", "alternate therapy prescribed") on different dates of service occurring on or after LOT1 start date	1,506
Patients with physician description of drug action in EHR	
Patients with at least 1 non-missing drug action (e.g., "start", "stop", "switch") occurring on or after LOT1 start date	2,003

Patients with at least 2 non-missing drug actions (e.g., "start", "stop", "switch") on difference dates of service occurring on or after LOT1 start date	1,730
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Appendix B. Characteristics of patients in Market Clarity

Demographics of patients in Market Clarity are presented below in Table 1. These counts were generated by Optum during the proposal process/development of the original start-up document. They are for information purposes only.

Table 1. Characteristics of Patients in Market Clarity and Population in US Census

Characteristic	Market Clarity ⁽¹⁾ (Jan 2007 – Jun 2022)		US Census Population ⁽²⁾ (2021)	
	Number of patients (million)	%	Number of people (million)	%
Total	76.8	100%	328.1	100%
Insurance Type				
Commercial	48.8	64%	216.4	66%
Medicaid	13.5	18%	61.9	19%
Medicare	13.4	17%	60.2	18%
Medicare Advantage + Part D	n/a	-	n/a	-
Other Payor Type	4.8	6%	3.2	1%
Uninsured	15.0	20%	27.2	8%
Unknown	10.1	13%	n/a	-
Census Region				
Midwest	30.4	40%	67.9	21%
Northeast	13.9	18%	56.1	17%
South	17.2	22%	126.1	38%
West	11.1	14%	77.9	24%
Unknown ⁽⁴⁾	4.1	5%	n/a	n/a
Age Cohort (years) ⁽³⁾				
0-18	10.2	13%	77.0	23%
19-25	5.6	7%	29.0	9%
26-34	9.3	12%	40.3	12%
35-54	20.2	26%	125.5	38%
55-64	10.6	14%	56.2	17%
65+	20.9	27%	56.2	17%
Unknown ⁽⁴⁾	0.0	0%	n/a	n/a
Gender				
Female	41.0	53%	166.2	51%
Male	35.8	47%	160.0	49%
Unknown	0.0	0%	n/a	n/a
Race/Ethnicity				
African American	7.4	10%	44.0	13%
Asian	1.8	2%	20.7	6%

Characteristic	Market Clarity ⁽¹⁾ (Jan 2007 – Jun 2022)		US Census Population ⁽²⁾ (2021)	
	Number of patients (million)	%	Number of people (million)	%
Caucasian	45.8	60%	194.2	59%
Hispanic	4.5	6%	62.5	19%
Other/Unknown	17.3	22%	6.7	2%

⁽¹⁾ The Optum[®] de-identified Market Clarity dataset
⁽²⁾ Census: Health Insurance Coverage in the United States 2022 Health Insurance Coverage in the United States 2021 (census.gov)
⁽³⁾ Age is calculated as 2022 minus year of birth
⁽⁴⁾ Unknown reflects instances when data fell outside of the defined range

Appendix C. Pressure-testing Patient Identification Criteria

Additional steps, summarized below, were implemented by Optum's team using the Optum Market Clarity data to confirm patients' MM diagnosis and to identify and exclude those with relapsed MM. These steps were supplemental to the inclusion and exclusion criteria and, ultimately, did not change the original sample size of N = 5,372 patients.

- **EHR flags for MM diagnosis date:** Patients meeting the following three criteria in the EHR diagnosis table were flagged: 1) diagnosis of ICD-9-CM=203.00 or ICD-10-CM code=C90.00; 2) diagnosis date +/- 30 days of the index date; 3) diagnosis status of "possible diagnosis of", "diagnosis of", "not recorded".
1. This step was used to confirm the claims-based diagnosis of newly diagnosed MM.
- **Claims flags for longer baseline periods:** Patients with each of 9, 12, and 18 months of baseline continuous enrollment were flagged. These longer baseline periods were considered to maximize identification of patients with MM patients before the identification period and determine the resulting sample size after these patients were excluded. GSK and Optum decided on a 6-month baseline period.
- **Claims flag for minimum 3-month follow-up period:** GSK and Optum explored a minimum 3-month follow-up period for all patients and decided on the minimum 6-month variable follow-up except for patients who died less than 6 months after the index date.

- **EHR flags for approved MM therapies:** Patients with ≥ 1 EHR record for approved MM therapies from the EHR prescriptions written, medications administered, and procedure tables were flagged. Approved MM therapies in the EHR tables were measured with the same NDC, CPT, and HCPCS codes applied to the claims data. These flags were used to identify the potential number of patients available for the EHR validation of the claims-based LOT.
- **Flags for baseline MM diagnoses:** Patients with ≥ 1 non-diagnostic medical claim for MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "possible diagnosis" of MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.
- **Exclusion of patients with baseline MM may have excluded those with smoldering MM, which can be a precursor to active MM:** A specific diagnosis code does not exist to differentiate smoldering from active MM. The implication of this exclusion is described in the limitations section.
- **Flags for baseline R/R MM:** Patients with ≥ 1 non-diagnostic medical claim for R/R MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "history of" of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.
- **EHR flags for other baseline cancers:** Patients with ≥ 2 EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified.
- **Flags for claims and EHR records with diagnoses for other cancers during follow-up:** Patients with ≥ 2 medical claims or EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified. These patients were identified because the LOT algorithm was developed for patients with MM only;

those with other cancers may have received different treatments. GSK and Optum decided not to exclude patients with other cancers in the follow-up period.

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